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Postmortem: Service disruption on January 21-22, 2020

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INSIDE FIGMA ENGINEERING

The root cause of our recent service outage and our next steps

TL;DR: Why did it happen?

Tuesday's incident was primarily caused by a long-running, expensive query. To prevent this issue from recurring, we will be improving our monitoring of expensive queries and setting tighter bounds on allowed running time.

The root cause of Wednesday's incident was that a routine change in database statistics caused PostgreSQL 9 to mis-plan the execution of one of our queries, causing expensive table scans and writes to temporary buffers. This was exacerbated by concurrent aggressive autovacuuming operations, which were happening because of a vacuuming backlog from the long-running query that had been terminated on Tuesday.

The upgrade to PostgreSQL 11 mitigated both of these causes: improvements to the query planner eliminated the possibility of the bad query plan, and the performance characteristics of autovacuuming have been significantly improved between versions 9 and 11.

https://www.figma.com/blog/post-mortem-service-disruption-on-january-21-22-2020/